



```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.holtwinters import SimpleExpSmoothing, ExponentialSmoothing
from sklearn.metrics import mean_squared_error, mean_absolute_error
from math import sqrt
```

```
In [2]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [10]: path = '/content/drive/MyDrive/Document/retail_store_inventory.csv'
```

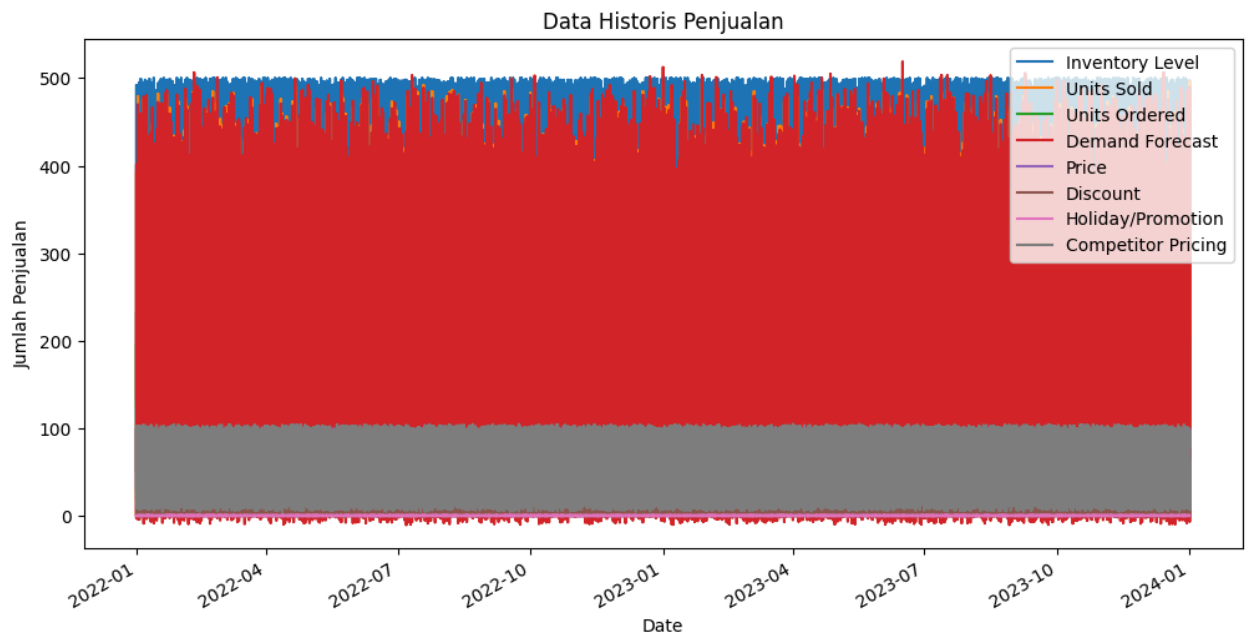
```
In [11]: df = pd.read_csv(path, index_col='Date', parse_dates=True)
```

```
In [13]: print("Tampilan Data Awal:")
display(df.head())
```

Tampilan Data Awal:

	Store ID	Product ID	Category	Region	Inventory Level	Units Sold	Units Ordered	Dema Foreca
Date								
2022-01-01	S001	P0001	Groceries	North	231	127	55	135.
2022-01-01	S001	P0002	Toys	South	204	150	66	144.
2022-01-01	S001	P0003	Toys	West	102	65	51	74.
2022-01-01	S001	P0004	Toys	North	469	61	164	62.
2022-01-01	S001	P0005	Electronics	East	166	14	135	9.

```
In [14]: df.plot(figsize=(12, 6), title="Data Historis Penjualan")
plt.ylabel("Jumlah Penjualan")
plt.show()
```



```
In [15]: target_col = 'Penjualan'
```

```
In [16]: train_size = int(len(df) * 0.8)
train = df.iloc[:train_size]
test = df.iloc[train_size:]
```

```
In [17]: print(f"Jumlah Data Latih: {len(train)}")
print(f"Jumlah Data Uji: {len(test)}")
```

Jumlah Data Latih: 58480

Jumlah Data Uji: 14620

```
In [23]: target_col = 'Units Sold'
```

```
ses_model = SimpleExpSmoothing(train[target_col]).fit(optimized=True)
test_predictions_ses = ses_model.forecast(len(test))
```

/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473:
ValueWarning: A date index has been provided, but it has no associated frequency information and so will be ignored when e.g. forecasting.

```
self._init_dates(dates, freq)
```

/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:837:
ValueWarning: No supported index is available. Prediction results will be given with an integer index beginning at `start`.

```
return get_prediction_index(
```

/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:837:
FutureWarning: No supported index is available. In the next version, calling this method in a model without a supported index will result in an exception.

```
return get_prediction_index(
```

```
In [24]: des_model = ExponentialSmoothing(train[target_col], trend='add', seasonal=None)
test_predictions_des = des_model.forecast(len(test))
```

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473:
ValueWarning: A date index has been provided, but it has no associated frequency
information and so will be ignored when e.g. forecasting.
    self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:837:
ValueWarning: No supported index is available. Prediction results will be given
with an integer index beginning at `start`.
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/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:837:
FutureWarning: No supported index is available. In the next version, calling this
method in a model without a supported index will result in an exception.
    return get_prediction_index(
```

```
In [25]: test['SES_Pred'] = test_predictions_ses.values
        test['DES_Pred'] = test_predictions_des.values
```

```
/tmp/ipykernel_5357/4143433640.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

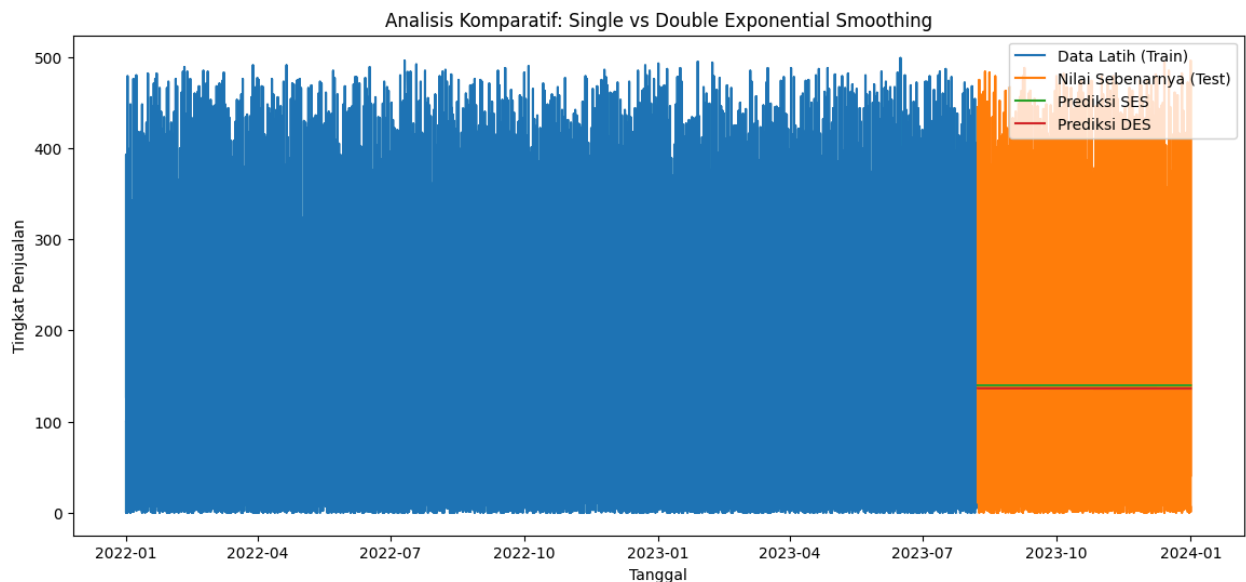
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
test['SES_Pred'] = test_predictions_ses.values
/tmp/ipykernel_5357/4143433640.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
test['DES_Pred'] = test_predictions_des.values
```

```
In [26]: plt.figure(figsize=(14, 6))
        plt.plot(train.index, train[target_col], label='Data Latih (Train)')
        plt.plot(test.index, test[target_col], label='Nilai Sebenarnya (Test)')
        plt.plot(test.index, test['SES_Pred'], label='Prediksi SES')
        plt.plot(test.index, test['DES_Pred'], label='Prediksi DES')
        plt.title("Analisis Komparatif: Single vs Double Exponential Smoothing")
        plt.xlabel("Tanggal")
        plt.ylabel("Tingkat Penjualan")
        plt.legend()
        plt.show()
```



```
In [27]: def evaluate_model(y_true, y_pred, model_name):
mae = mean_absolute_error(y_true, y_pred)
rmse = sqrt(mean_squared_error(y_true, y_pred))
print(f"--- Performa {model_name} ---")
print(f"MAE : {mae:.2f}")
print(f"RMSE : {rmse:.2f}\n")
```

```
In [28]: evaluate_model(test[target_col], test['SES_Pred'], "Single Exponential Smoothing")
evaluate_model(test[target_col], test['DES_Pred'], "Double Exponential Smoothing")
```

```
--- Performa Single Exponential Smoothing ---
MAE : 89.42
RMSE : 108.35
```

```
--- Performa Double Exponential Smoothing ---
MAE : 88.78
RMSE : 108.28
```

```
In [29]: best_rmse = min(sqrt(mean_squared_error(test[target_col], test['SES_Pred'])),
sqrt(mean_squared_error(test[target_col], test['DES_Pred'])))
```

```
In [30]: print("--- Kesimpulan Manajerial ---")
print(f"Error prediksi (RMSE) terendah adalah sekitar {best_rmse:.2f} unit.")
print("Dalam konteks Manajemen Persediaan, nilai RMSE ini dapat dijadikan estimasi standar deviasi error peramalan.")
print(f"Untuk mencegah kehabisan stok (stockout), manajemen disarankan untuk menyiapkan Safety Stock tambahan proporsional dengan angka 108.28 unit ini, dikalikan dengan Z-score dari *Service Level* yang diinginkan.")
```

```
--- Kesimpulan Manajerial ---
Error prediksi (RMSE) terendah adalah sekitar 108.28 unit.
Dalam konteks Manajemen Persediaan, nilai RMSE ini dapat dijadikan estimasi standar deviasi error peramalan.
Untuk mencegah kehabisan stok (stockout), manajemen disarankan untuk menyiapkan Safety Stock tambahan proporsional dengan angka 108.28 unit ini, dikalikan dengan Z-score dari *Service Level* yang diinginkan.
```